# *Abhinav Goel*

*Residence/Domicile:* Muzaffarnagar, Uttar Pradesh, India - 251001
*E-mail:* abhinav.goel.robotics@gmail.com ✻ *Phone number:* +91-8384857368
*Place of birth:* Muzaffarnagar, Uttar Pradesh, India ✻ *Date of birth:* 18-12-2004
*LinkedIn:* Abhinav Goel ✻ *GitHub:* Humobot1812 ✻ *Portfolio:* Humobot1812.github.io/Portfolio

## *Education*

**B. Tech. in Electronics and Communication Engineering** — *PDPM IIITDMJ*
*Bachelor's Degree Program* — *August 2024 - Present*
*Semester IV (CPI - 8.5)*

**Senior Secondary (Higher Secondary)** — *PR Public School, Muzaffarnagar*
*Central Board of Secondary Education (CBSE)* — *July 2022 - July 2024*
*Percentage: 93.6%*

**Secondary (High School)** — *Shardein School, Muzaffarnagar*
*Central Board of Secondary Education (CBSE)* — *April 2020 - July 2022*
*Percentage: 98.4%*

## *Areas of interest*

- Autonomous Mobile Robots (AMRs), SLAM, Navigation and Trajectory Planning (ROS 2, Nav2)
- Unmanned Aerial Vehicles (UAVs) & Autonomous Aerial Flight Systems
- Robotic Manipulation, Subterranean, Field Exploration & Bio-Inspired Legged Robotics
- Computer Vision, Spatial Geolocation & Multi-Spectral Perception & ORB SLAM
- Robotics Programming
- Internet of things (IoT)
- Robotics and IoT system design

## *Honors and key achievements*

- **e-Yantra Robotics Competition (e-YRC 2025–26), IIT Bombay:** Top 10 National Finalist among 400+ participating teams across India in the *KrishiCobot* theme, designing an autonomous agricultural harvesting mobile manipulator with ROS 2, UR5 visual servoing, and precision vision pipelines. [Certificate]
- **DD Robocon 2025 (Doordarshan & IIT Delhi):** Achieved Stage-2 National Qualification representing Team Barbarick with an evaluation score of 86/100, designing competition-grade holonomic robot chassis and actuation mechanisms. [Certificate]
- **SAE Aerothon 2025 (SAE India):** Phase 2 National Qualification in the premier aerial robotics challenge, demonstrating an autonomous fixed-altitude surveillance quadcopter with ground target pixel-to-GPS conversion. [Certificate]
- **NIDAR 2026 (National Innovation Challenge for Drone Application and Research):** National Finalist, Aero Fabrication Club, developing an autonomous disaster surveillance UAV with 75+ acre automated polygon coverage and real-time survivor detection. [Certificate]

## *Research and development experience*

**Autonomous Agricultural Multi-Robot System (KrishiCobot)** *August 2025 - May 2026*

*e-Yantra Robotics Competition 2025–26, IIT Bombay* *github.com/Humobot1812/Krishi_Cobot*

- Developed a multi-robot agricultural system integrating an autonomous mobile robot (eBot) for greenhouse navigation with a dock-based UR5 manipulator for automated fruit and fertilizer handling in Gazebo.
- Implemented a LiDAR-based RANSAC shape detector converting polar scans to Cartesian clusters, fitting lines and verifying vertex angles to classify triangular (fertilizer) and square (bad health) plant markers.
- Engineered a vision pipeline utilizing ArUco markers, HSV color segmentation, contour centroid extraction, and IQR depth filtering, broadcasting dynamic TF2 transforms for targets and docking stations.
- Programmed UR5 delta-twist servo control and PID positioning with intermediate waypoints to avoid kinematic singularities; qualified as a Top 10 National Finalist among 400+ collegiate teams across India.

**Autonomous Warehouse Stock Auditing AMR & Active Vision System (WareOps)** *December 2025 - August 2026*

*Robotics & Autonomous Systems, PDPM IIITDMJ* *github.com/WareOps*

- Architected a distributed ROS 2 Humble AMR system splitting workloads over Wi-Fi between an onboard Raspberry Pi 4 (navigation/control) and a remote operator workstation (vision/logging/teleop).
- Developed a custom C++ ros2_control hardware interface plugin (diffdrive_arduino) over USB serial with an Arduino Nano running ROSArduinoBridge firmware for motor PID control and quadrature encoder feedback.
- Configured SLAM Toolbox for 2D mapping and Nav2 with AMCL for autonomous rack-to-rack path planning and obstacle avoidance using an RPLidar A1M8 in physical warehouse environments.
- Engineered an active-vision scanning mast with an ESP32-CAM Wi-Fi MJPEG stream and HTTP pan-tilt servo tracking, running OpenCV WeChat QR and PyZbar to decode rack barcodes and log records to Excel and CSV ledgers.

**CAVE DIVER — Autonomous 6-Wheeled Subterranean Exploration Rover** *September 2026 - Present*

*Smart India Hackathon 2026, PDPM IIITDMJ* *github.com/Humobot1812/CAVE_DIVER_6_Wheeled_Rover*

- Engineered an autonomous 6-wheeled robotic rover platform in Ignition Gazebo (Fortress) with custom Armo_bot URDF and cave SDF worlds for GPS-denied subterranean reconnaissance and hazard monitoring.
- Integrated dual-spectrum visual inspection, capturing synchronized optical RGB (/camera/image_raw) and LWIR thermal (/thermal/image_raw) video feeds with tactical HUD overlays for survivor and obstacle detection.
- Developed an atmospheric hazard telemetry suite continuously monitoring six environmental parameters: Temperature, Relative Humidity, $CO_2$, $O_2$, Methane ($CH_4$), and Carbon Monoxide (CO).
- Built a web Ground Control Station in Flask 3.1 & HTML5/WebSocket (port 5001) rendering live dual feeds, gas dials, a 2D LiDAR radar sweep display, spotlight control, and QR matrix landmark navigation via ros_gz_bridge.

**Tactical Multi-UAV Ground Control Station (DroneGCS)** *September 2026 - Present*

*Aero Fabrication Club, PDPM IIITDMJ* *github.com/Humobot1812/DroneGCS*

- Developed a tactical Ground Control Station in C++17, Qt 6, and QML for multi-rotor, fixed-wing, and VTOL drones communicating via MAVLink v2 with ArduPilot and PX4 autopilots over UDP, TCP, and serial.
- Integrated a native Tactical AI Copilot powered by Google Gemini AI, enabling natural language flight commands, automated RTL execution, and conversational telemetry SITREP briefings.
- Built a Primary Flight Display (PFD) HUD with artificial horizon, Google Satellite imagery, and a live multi-channel Telemetry Oscilloscope plotting altitude, climb rate, battery voltage, and groundspeed.
- Engineered a Geofence Safety Console for setting cylindrical boundaries, altitude ceilings, and keep-in/out corridors, with automated synchronization from flight controllers and low-latency GStreamer H.264 video.

**SafePilot — Intelligent Flight Assistance & Safety Envelope System** *September 2026 - Present*

*Open-Source UAV Development* *github.com/Humobot1812/SafePilot*

- Created an open-source ROS 2 Humble package (safe_pilot) wrapping MAVSDK to provide beginner-safe flight assistance and software-enforced safety envelopes for ArduPilot and PX4 drones.
- Implemented 3D body-frame velocity teleoperation via gamepad/joystick with single-button discrete flight commands (Arm, Takeoff, Land, RTL, Hold, Kill) and live speed scaling via D-pad.
- Engineered a multi-core AI hand gesture control worker process with CPU affinity pinning and MediaPipe Hand Landmarker (TFLite XNNPACK delegate) for touchless strafe, climb, and closed-fist yaw rotation.
- Configured embedded polygon geofencing with altitude ceilings triggering automated auto-RTL on boundary breach, with one-command unified Gazebo Harmonic and ArduPilot SITL simulation launch.

**Autonomous Disaster Surveillance UAV & Victim Detection** *June 2025 - January 2026*

*NIDAR 2026 National UAV Challenge, Aero Fabrication Club* *github.com/Humobot1812/Autonomous_drone*

- Developed an autonomous aerial search and rescue platform for the NIDAR 2026 National UAV Challenge, integrating computer vision, GPS mapping, and autonomous navigation using MAVSDK Python.
- Implemented an autonomous survey coverage system parsing .kml polygon boundary files to systematically sweep large disaster zones at 25 m altitude.
- Deployed real-time AI human detection models on live downward camera feeds, converting pixel detections to geographic GPS coordinates and dynamically controlling camera pan/tilt tracking toward victims.
- Validated complete mission execution (automated arming, takeoff, waypoint traversal, victim loiter, and RTL) across ArduPilot SITL simulation and physical multirotor hardware flight tests.

**Autonomous Aerial Survey & Ground Target Geolocation Drone** *May 2025 - November 2025*

*SAE Aerothon 2025, Aero Fabrication Club, PDPM IIITDMJ* *github.com/Humobot1812/Aerothon-25*

- Engineered an autonomous survey and object detection drone using ROS 2, DroneKit, and Pixhawk for competitive presentation at SAE Aerothon 2025.
- Deployed YOLOv4-tiny using OpenCV DNN on an onboard companion computer for real-time edge target and hotspot identification during autonomous flight.
- Designed an automated mission interrupt pipeline: switches from AUTO to GUIDED mode upon target detection, descends, performs PID-based visual centering and hover inspection, then climbs and resumes survey.
- Derived an image pixel-to-GPS conversion algorithm considering camera FOV, altitude, and heading, applying Haversine spatial filtering (<0.8 m) to eliminate duplicate hotspot coordinate records.

**Differential Drive Mobile Robot — ROS 2 Navigation & SLAM** *June 2025 - July 2025*

*Electronics and Robotics Society, PDPM IIITDMJ* *github.com/Humobot1812/Differential-Drive-ROS2-Navigation-Project*

- Constructed a complete simulation and navigation stack for a 4-wheeled differential-drive robot (Armo_description) using ROS 2 Humble and Gazebo Harmonic.
- Modeled kinematic chains, wheel inertia matrices, collision geometries, 2D LiDAR, and IMU plugins using modular URDF/Xacro macros with gazebo_bridge integration.
- Integrated robot_localization Extended Kalman Filter (EKF) fusing wheel odometry and IMU data to produce drift-free robot state estimation in complex maze environments.
- Built 2D occupancy grid maps using SLAM Toolbox and configured Nav2 with AMCL particle filtering for autonomous obstacle avoidance, global path planning, and goal pose tracking.

**Smart Multi-Protocol ESP32 Robotics Transmitter System** *April 2026 - July 2026*

*Electronics and Robotics Society, PDPM IIITDMJ* *github.com/Humobot1812/Smart-Multi-Protocol-ESP32-Transmitter*

- Designed and built a modular smart transmitter utilizing a dual-ESP32 architecture: a Controller ESP32 reading inputs and a Dashboard ESP32 managing UI and RF transmission via UART.
- Interfaced 2 analog joysticks, 5 potentiometers, a 4×4 matrix keypad, and digital push buttons for comprehensive multichannel robotics teleoperation.
- Programmed a 128×64 ST7920 graphical LCD using the U8g2 library, featuring boot animations, keypad password security, real-time channel bar graphs, battery monitoring, and wheel ratio configuration.
- Supported four switchable wireless protocols (NRF24L01+ 2.4 GHz, ESP-NOW 250 m, Wi-Fi UDP, and Bluetooth), with an onboard MPU6050 IMU enabling tilt-based gesture control.

**Custom ESP32-S3 Quadcopter Drone with Discrete MOSFET ESCs** *January 2026 - February 2026*

*Aero Fabrication Club, PDPM IIITDMJ* *youtube.com/playlist?list=PLQcgql__dXrcVR0A4IEijRmWpTvdtAOpG*

- Designed and fabricated custom 3-phase Electronic Speed Controllers (ESCs) using discrete N-channel MOSFETs and gate drivers, eliminating commercial off-the-shelf ESC modules.
- Implemented 6-step trapezoidal commutation firmware on an ESP32-S3 microcontroller, achieving smooth BLDC motor speed regulation with configurable acceleration profiles.
- Programmed a 500 Hz flight stabilization PID loop sampling angular velocities and accelerations from an MPU6050 6-axis IMU for real-time roll, pitch, and yaw stabilization.
- Integrated a BMP280 barometric pressure sensor for relative altitude estimation and hover stabilization, validating performance through physical motor bench testing and flight trials.

**5-DOF Intelligent Robotic Arm with AI Hand Tracking & ESP32-CAM** *March 2025 - Present*

*Electronics and Robotics Society, PDPM IIITDMJ* *github.com/Humobot1812/Robotic_arm*

- Fabricated an articulated 5-DOF robotic arm (Base, Shoulder, Elbow, Wrist, Gripper) driven by an Arduino Uno interfacing with a PCA9685 16-channel I²C servo controller.
- Mounted an ESP32-CAM on the end-effector providing real-time wireless Wi-Fi video streaming of manipulation tasks to a custom desktop operator dashboard.
- Developed an AI vision teleoperation pipeline in Python using OpenCV and MediaPipe, tracking 21 hand landmarks to convert human hand motion into real-time arm servo commands.
- Implemented pinch gesture recognition for gripper open/close and geometric Forward and Inverse Kinematics (FK/IK) with workspace boundary checks and KP gain tuning over USB serial.

**Bio-Inspired Serpentine Snake Robot Simulation** *June 2025 - July 2025*

*Electronics and Robotics Society, PDPM IIITDMJ* *github.com/Humobot1812/Snake_Robot*

- Engineered a bio-inspired multi-segment snake robot simulation in ROS 2 Humble and Gazebo utilizing modular snake_robot_description and snake_robot_controller packages.
- Constructed modular body segments interconnected with revolute joints in URDF/Xacro, configuring Gazebo physics plugins and ros_gz_bridge topic communications.
- Developed a custom Python controller node generating traveling sinusoidal undulation waves by publishing coordinated phase-shifted joint position commands to actuate body links.
- Implemented configurable motion parameters allowing real-time adjustment of wave amplitude, frequency, and inter-joint phase offsets for smooth snake-like planar locomotion.

**IoT Hostel Attendance & Access Control System using ESP8266 & RFID** *October 2025 - November 2025*

*Campus Automation Project, PDPM IIITDMJ* *github.com/Humobot1812/Hostel_Attendance_system*

- Engineered an IoT-based hostel attendance system utilizing an ESP8266 (NodeMCU) and an MFRC522 RFID reader interfaced via SPI with MIFARE 1K RFID cards.
- Programmed automatic entry/exit state machine logic: initial RFID scan marks Time In and subsequent scan marks Time Out, tracking Student ID and gate numbers automatically.
- Integrated direct cloud logging: ESP8266 transmits HTTPS POST requests over Wi-Fi to a Google Apps Script Web App (sheet.gs), updating records in real-time Google Sheets without dedicated local servers.
- Developed companion firmware (rfid_tag.ino) for writing student credentials onto RFID tags; deployed and validated system functionality in institute hostel operations.

## *Certifications*

- **e-Yantra Robotics Competition (e-YRC 2025–26) Finalist:** Awarded by IIT Bombay for securing Top 10 nationally among 400+ collegiate teams in the autonomous KrishiCobot agricultural robotics theme (August 2025 – May 2026).
- **DD Robocon 2025 Stage-2 Certificate:** Awarded by Doordarshan & IIT Delhi for qualifying Stage 2 with Team Barbarick, scoring 86/100 in technical design and mechanism evaluations (2025).
- **SAE Aerothon 2025 National Aerial Robotics Certificate:** Awarded by SAE India for qualifying Phase 2 in the national autonomous UAV surveillance challenge (June 2025).
- **NIDAR 2026 National UAV Challenge:** Certified participant in National Innovation Challenge for Drone Application and Research (January 2026).
- **Hands-on Introduction to Autonomous Systems with ROS 2 & Embedded Hardware:** PDPM IIITDMJ Student Technical Workshop (2024).

## *Technical Skills*

| | |
|---|---|
| **Robotics & Autonomous Systems:** | ROS 2 (Humble), Nav2, ros2_control, SLAM Toolbox, AMCL, TF2, URDF/Xacro, MoveIt 2, Kinematics, Trajectory Planning, Sensor Fusion, EKF, RANSAC |
| **UAV & Flight Systems:** | ArduPilot, PX4, Pixhawk, MAVLink, MAVSDK, DroneKit, QGroundControl, Mission Planner, ArduPilot SITL, PID Control |
| **Computer Vision & Perception:** | OpenCV, YOLOv8, YOLOv4-Tiny, ArUco, Visual Servoing, Pinhole Camera Model, ArUco Marker Geolocation, ORB-SLAM3 |
| **Embedded Systems & Electronics:** | ESP32-S3, ESP32-CAM, Arduino, STM32, UART, I²C, SPI, PWM, MPU9250/MPU6050, Encoder Motors, Barometer, NRF24L01, LoRa, ST7920 Display, Custom MOSFET ESC |
| **Compute, Sensors & Hardware:** | Raspberry Pi 4/5, NVIDIA Jetson Nano/AGX Xavier, RPLidar A1M8, LiDAR, Raspberry Pi Camera v3 NoIR |
| **Simulation & Engineering Tools:** | Gazebo, RViz2, MuJoCo, MATLAB/Simulink, Fusion 360, 3D Printing, TinkerCAD, Git, Linux (Ubuntu 22.04) |
| **Programming Languages:** | Python, C++ |

## *Language proficiency*

| | |
|---|---|
| **English, Hindi** | Full Professional Proficiency |

## *Memberships*

- **Electronics and Robotics Society of PDPM IIITDMJ** (Coordinator, March 2026 – Present | Member, August 2024 – March 2026)
- **Aero Fabrication Club of PDPM IIITDMJ** (Core Committee Member, May 2025 – Present | Member, August 2024 – May 2025)